ACADEMIC CENTRAL

PRACTICE EXAMINATION

N6

INDUSTRIAL ELECTRONICS N6

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central N6 Industrial Electronics N6 paper 1 · practice a.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct methods and clearly labelled sketches.

QUESTION 1

1.1 Circuit: three input resistors to the inverting input, R_f from output to inverting input, non-inverting input to 0 V ✓; $V_o = -100\text{ k}(\frac{0{,}5}{20\text{ k}} + \frac{-1}{50\text{ k}} + \frac{0{,}2}{25\text{ k}})$ ✓ $= -(2{,}5 - 2 + 0{,}8)$ ✓✓ $= -1{,}3$ V ✓ (5)

1.2.1 $\frac{\Delta V_o}{\Delta t} = -\frac{V_{in}}{RC} = -\frac{2}{10 \times 10^3 \times 0{,}1 \times 10^{-6}}$ ✓✓ $= -2\,000$ V/s ✓ (3)

1.2.2 $V_o = -2\,000 \times 0{,}002$ ✓ $= -4$ V (within the supply limits) ✓ (2)

1.3 Triangular input drawn ✓; output is a square wave ✓; output negative while the input rises and positive while the input falls (inverting) ✓; transitions line up with the peaks of the triangle ✓ (4)

1.4.1 $A_v = 1 + \frac{90}{10}$ ✓ $= 10$ ✓ (2)

1.4.2 $V_o = 10 \times 0{,}4$ ✓ $= 4$ V ✓ (2)

1.4.3 About +13,5 V ✓ – 10 × 2 = 20 V is more than the supply allows, so the output saturates ✓ (2)

[20]

QUESTION 2

2.1 Axes labelled V_{AK} and I_A ✓; forward blocking region rising to the forward breakover voltage V_{BO} ✓; sudden switch to the low-voltage conducting region ✓; holding current I_H marked ✓; reverse blocking region with reverse breakdown ✓ (5)

2.2.1 $V_m = \sqrt{2} \times 230 = 325{,}27$ V ✓; $V_{av} = \frac{325{,}27}{2\pi}(1 + \cos 60°)$ ✓✓ $= 77{,}65$ V ✓ (4)

2.2.2 $I_{av} = \frac{77{,}65}{20}$ ✓ $= 3{,}88$ A ✓ (2)

2.3 $V_{av} = \frac{2V_m}{\pi}\cos\alpha$ ✓ $= \frac{2 \times 325{,}27}{\pi} \times \cos 30°$ ✓✓ $= 179{,}33$ V ✓ (4)

2.4 An SCR conducts in one direction only (unidirectional) ✓; a triac conducts in both directions and can be triggered in either half-cycle (bidirectional) ✓, so it controls AC power ✓✓; application: light dimmer / AC motor speed control / heater control ✓ (5)

[20]

QUESTION 3

3.1.1 Transformer ✓; four diodes in a bridge, correctly orientated ✓; filter capacitor across the load with polarity ✓ (3)

3.1.2 $V_p = \sqrt{2} \times 12 - 2 \times 0{,}7$ ✓ $= 16{,}97 - 1{,}4 = 15{,}57$ V ✓ (2)

3.1.3 $V_r = \dfrac{0{,}5}{100 \times 2\,200 \times 10^{-6}}$ ✓ $= 2{,}27$ V ✓ (2)

3.1.4 $V_{DC} \approx 15{,}57 - \dfrac{2{,}27}{2} = 14{,}43$ V ✓ (1)

3.2.1 $I_L = \dfrac{9{,}1}{200}$ ✓ $= 45{,}5$ mA ✓ (2)

3.2.2 $R_s = \dfrac{15 - 9{,}1}{0{,}0455 + 0{,}005}$ ✓✓ $= 116{,}83\ \Omega$ ✓ (3)

3.2.3 $I_z = \dfrac{5{,}9}{116{,}83} = 50{,}5$ mA ✓; $P_z = 9{,}1 \times 0{,}0505$ ✓ $= 0{,}46$ W ✓ (3)

3.3 Advantages (any TWO) ✓✓: high efficiency (less heat); smaller and lighter; wide input voltage range. Disadvantages (any TWO) ✓✓: more complex and expensive; electrical noise / interference (EMI); more output ripple (4)

[20]

QUESTION 4

4.1.1 $f = \dfrac{1}{2\pi \times 10 \times 10^3 \times 10 \times 10^{-9}}$ ✓✓ $= 1\,591{,}55$ Hz ✓ (3)

4.1.2 The amplifier gain must be 3 (the network attenuates to ⅓ at resonance) ✓✓; 1 + R_f/R_1 = 3, so $R_f = 2R_1$ ✓ (3)

4.2.1 $t_H = 0{,}693 \times 11 \times 10^3 \times 10 \times 10^{-6}$ ✓ $= 76{,}23$ ms ✓; $t_L = 0{,}693 \times 10 \times 10^3 \times 10 \times 10^{-6}$ ✓ $= 69{,}3$ ms ✓ (4)

4.2.2 $T = 145{,}53$ ms ✓; $f = \dfrac{1}{0{,}14553} = 6{,}87$ Hz ✓; duty cycle $= \dfrac{76{,}23}{145{,}53} \times 100$ ✓ $= 52{,}38\%$ ✓ (4)

4.3 The loop gain Aβ must be equal to (at least) 1 ✓, and the total phase shift around the loop must be 0° or 360° (positive feedback) ✓ (2)

4.4 Any TWO: very stable frequency ✓; high accuracy / not affected much by temperature and supply changes ✓; small and cheap (2)

4.5 Any TWO: time-delay relays ✓; flashing warning lights ✓; PWM motor speed control; pulse generators; debouncing switches (2)

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QUESTION 5

5.1.1 $\Delta R = GF \times R \times \varepsilon$ ✓ $= 2 \times 120 \times 500 \times 10^{-6}$ ✓✓ $= 0{,}12\ \Omega$ ✓ (4)

5.1.2 $V_o \approx \frac{V \cdot GF \cdot \varepsilon}{4}$ ✓ $= \frac{10 \times 2 \times 500 \times 10^{-6}}{4}$ ✓✓ $= 2{,}5\ \text{mV}$ ✓ (4)

5.2 Thermocouple – two different metals produce a small voltage proportional to the temperature difference (Seebeck effect) ✓; advantage: very wide range, robust, cheap ✓. RTD – resistance of a metal (e.g. platinum) increases almost linearly with temperature ✓; advantage: accurate, stable, linear ✓. Thermistor – semiconductor whose resistance changes greatly with temperature (NTC or PTC) ✓; advantage: very sensitive, cheap, small ✓ (6)

5.3 Sketch: primary coil in the middle, two secondary coils on either side connected in series opposition, movable iron core ✓; AC on the primary induces voltages in the secondaries; at the centre they cancel (zero output) ✓; when the core moves, one secondary voltage increases and the other decreases, giving an output proportional to displacement, with phase showing direction ✓ (3)

5.4 An active transducer generates its own electrical output without an external supply ✓, e.g. thermocouple, piezoelectric crystal, solar cell; a passive transducer needs an external supply because it changes a property such as resistance ✓, e.g. strain gauge, LDR, RTD (one correct example of each ✓) (3)

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TOTAL: 100

END OF MARKING GUIDELINE